

US Visa Certification — Classification Model

Data Science & Business Analytics project · Adewale Adeyinka Falade

Problem

US employers review large volumes of work-visa applications each year. The goal: a classification model that predicts whether an application will be **Certified** or **Denied**, and surfaces the factors that most influence the outcome — to speed shortlisting and support evidence-based decisions.

Data & approach

25,480 applications with 12 features (continent, education, job experience, prevailing wage, region, employer size, full-time status, etc.); target is class-imbalanced (~67% certified). Workflow: categorical encoding and data-quality checks → EDA → handling class imbalance (over- and under-sampling, balanced class weights) → ensemble models (Random Forest, AdaBoost, Gradient Boosting, XGBoost) → hyperparameter tuning with GridSearchCV, selecting on **F1** (which balances wrongly certifying vs. wrongly denying a qualified applicant).

Results — tuned models

Model	Accuracy	Recall	Precision	F1
Random Forest	0.777	0.911	0.788	0.845
AdaBoost	0.749	0.876	0.777	0.823
Gradient Boosting	0.750	0.899	0.767	0.828
XGBoost (selected)	0.762	0.980	0.745	0.846

XGBoost gave the best balance (highest recall and F1); Random Forest was a close competitor. Boosting methods generalised well between training and validation, while bagging/Random Forest showed more overfitting before tuning.

Key drivers of certification

Higher education level (Doctorate > Master's > Bachelor's), prior job experience, applicant continent (Europe and Africa higher), and prevailing-wage band.

Recommendation

Deploy the tuned XGBoost model for certification prediction (use Gradient Boosting where an under-sampled training regime is preferred), paired with the driver analysis to guide applicant shortlisting.

Tools: Python · pandas · NumPy · scikit-learn · XGBoost · matplotlib · seaborn. Full analysis notebook is included in this repository.